MIL-I-55684(EL)

6 July 1970

MILITARY SPECIFICATION

INSTALLATION AND ACCEPTANCE TESTING OF

ELECTRONIC ALTIMETER SET AN/APN-171A(V)

1. SCOPE

1.1 Scope. This specification provides installations, equipment performance, engineering and production approval requirements for installation and testing of Electronic Altimeter Set AN/APN-171A(V)1 and associated equipment in prototype and production model aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-17	Cables, Radio Frequency; Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead
MIL-B-5087	Bonding, Electrical, and Lightning Protection for Aerospace Systems
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-I-6051	Interference Limits and Methods of Measurements, Electronic Installation in Airborne Weapons Systems and Associated Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipments

FSC 5841

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MIL-I-8700	Installation and Test of Electronic Equipment in Aircraft, General Specification for
MIL-C-23329	Connectors, Coaxial, Radio Frequency, Series BNC, TNC, N, C and SC
MIL-C-45662	Calibration Equipment, Aircraft, General Specification for
MS91405-A1B	Base, Mounting, Large Size (for use with Electronic Equipment in Aircraft)

STANDARD

Military

MIL-STD-704	Electrical Power, Aircraft, Characteristics and Utilization of
MIL-STD-877	Antenna Subsystem, Airborne, Criteria For Design and Location of

PUBLICATIONS

Technical Manual

NAVAIR 16-30 APN171-1	Electronic Altimeter Set, AN/APN-171(V)
NAVAIR 16-45-1205-1	Radar Altimeter Test Set, UG1984B1 (Honeywell Inc.)

3. REQUIREMENTS

3.1 General

3.1.1 Acceptance. Any installation of Electronic Altimeter Set sub-system which is in accordance with MIL-I-8700 and the requirements of this specification shall be considered acceptable. Such acceptance, however, does not constitute safety-of-flight approval.

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3.1.2 Responsibility for Miscellaneous Components. The contractor shall furnish and install all required cables, connectors, terminal boards, junction boxes, circuit breakers, and miscellaneous items to complete satisfactory installation and operation of the Electronic Altimeter subsystem. Interconnecting cables are not furnished as part of the Electronic Altimeter sub-system. All contractor furnished electronic equipment shall be in accordance with MIL-I-6181 and MIL-E-5400. The requirements in MIL-E-5400 for Class 2 equipment designed for operation in both aircraft and helicopters shall apply.

3.1.3 Quality Assurance. The contractor shall perform all examinations and tests specified in Section 4 to determine compliance with the applicable requirements of Section 3.

3.1.4 Reports. The installing activity shall prepare and submit to the procuring activity all reports and other supplementary data as specified herein.

3.1.5 Check-off List. Results of the applicable examinations and test as specified in Section 4 shall be recorded on check-off lists similar to the examples shown in Appendix A. A check-off list shall be prepared for each Electronic Altimeter sub-system installed in an aircraft and a file maintained of these records for reference, as necessary, by the procuring activity.

3.1.6 Equipment Failure. In case the equipment fails to comply with the requirements of 3.3, the requirements and procedures as specified in MIL-I-8700 shall govern with the exception that replacement of component parts shall be made only upon specific approval by the procuring activity. AMC Form 1229 "Unserviceable New Materiel Report" shall be utilized when reporting equipment failure. Shipping instructions for the faulty equipment and distribution requirements for the equipment failure report shall be specified by the contracting officer. A sample of AMC Form 1229 is shown in Figure 1.

3.1.7 Test Equipment Calibration. The contractor shall prepare a statement which identifies the test equipment utilized in the performance of the tests specified in section 4. The statement shall include certification of the test equipment calibration in accordance with MIL-C-45662 and be forwarded to the procuring activity as supplementary data to the test report as specified herein.

3.1.8 Electronic Sub-system Certification. The contractor shall prepare a statement which certifies that all electronic sub-systems installed in the aircraft or otherwise utilized in the performance of the tests as specified in Section 4, are in accordance with the applicable specifications. The statement shall be forwarded to the procuring activity as supplementary data to the applicable test reports as specified herein.

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3.1.9 Procuring Activity Approval. In all cases where approval by the procuring activity is specified, the contractor shall be in receipt of said approval prior to conducting any tests, inspections or otherwise utilizing the proposed equipment or procedures.

3.1.10 Supplementary Data. All supplementary data as specified herein shall be prepared by the contractor and forwarded to the procuring activity as specified herein.

3.1.10.1 Instrumentation. The instrumentation system application and techniques utilized to collect the engineering flight test data shall be prepared and forwarded to the procuring activity as supplementary data to the engineering flight test report. The range and accuracy (including transducer and print-out errors of each recorded signal) shall be adequate for comprehensive evaluation of the parameter(s) being measured. The supplementary data shall include, but not be limited to, the following:

a. Recording media (block diagram including component identification numbers).

b. Transducer and/or signal source parameters.

c. Signal conditioning for each signal (schematic).

d. Preflight and post flight calibration techniques.

e. Calibration curves necessary for complete analysis of each recorded signal.

f. Other data as required.

3.1.11 Electromagnetic Compatibility. Interference control and test plans shall be prepared in accordance with MIL-E-6051 and submitted to the procuring activity for approval. After receipt of approval, the contractor shall conduct the tests as specified in 4.3 to demonstrate compliance with MIL-E-6051 and to insure interference free operation of the Electronic Altimeter sub-system after installation in an aircraft. An electromagnetic compatibility test report shall be forwarded to the procuring activity as directed by the contracting officer.

3.1.12 Equipment to be Tested. The equipment listed in Table I shall meet the requirements as specified herein. Whenever the model or detail specification for a particular aircraft provides a list not in agreement with Table I, the list in the aircraft specification shall govern.

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TABLE I. Electronic Equipment to be Tested for Electronic Altimeter Set

EQUIPMENT DESIGNATION	NOMENCLATURE AND PART NUMBER	NO. OF UNITS REQUIRED	WT. PER UNIT (POUNDS)	UNIT HEAT DISSIPATION (MAX.)
RT-804A/ APN-171(V)	Receiver-Transmitter HG9050E1	1	13.4	80W
ID-1345A/APN-171A(V)	Height Indicator JG206D5	1	1.5	15W
F-1263/APN-171A(V)1	Radio Interference Filter LG1026AA01	1	0.25 Max.	None

3.1.13 Additional Equipment Required. In addition to the equipment listed in TABLE I, the following equipment, listed in TABLE II, is required for a complete installation of the Electronic Altimeter Sub-system.

TABLE II. Electronic Equipment Assembly Table for Electronic Altimeter Set

EQUIPMENT DESIGNATION	NOMENCLATURE AND PART NUMBER	NO. OF UNITS REQUIRED	WT. PER UNIT (POUNDS)
AS-1858/APN-171(V)	Antenna LG81A4	2	1
MT-4167/APN-171(V)	Mounting Base WG1000AA01	1	2.2 Max.

3.1.14 Test Equipment Required. The following test equipment listed in Table III, or equipment of equal or superior characteristics, is recommended for use in the performance of the tests specified herein. Items which are considered unique to the equipment under test and which may be loaned to the aircraft contractor by the Electronics Command are marked with an asterisk(*). Other items may be supplied in special circumstances, but in all cases such circumstances must be justified by the contractor and approved by the Electronics Command.

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TABLE III. Test Equipment and Accessories

NAME	PREFERRED TYPE	
	TYPE DESIGNATION	CHARACTERISTICS
Differential Voltmeter	ME-202/U	0.1 to ±500 vdc ± .05%, 8 steps 0.01 to 500 vdc ± .02%
Radar Altimeter System Tester *	Honeywell HG2308AA01	Shop level test set Used to check, calibrate and maintain AN/APN-171A(V)1 Electronic Altimeter Set
Multimeter	TS-352 B/U	ac and dc voltage measuring capability to 150V ohmmeter function for checking continuity
Zero Calibration Cable	See 4.5.2	Simulate aircraft installation RF delays
Stop Watch		Range: 0.1 second to 5.0 minutes

3.2 Installation (See 4.4). All Government furnished aircraft equipment and contractor furnished equipment shall be installed in accordance with MIL-I-8700 and as specified herein.

3.2.1 Electrical Wiring. All electrical wiring required for installation of the Electronic Altimeter Set in an aircraft shall be in accordance with MIL-W-5088 and as specified herein.

3.2.2 Electrical Bonding. Electrical Bonding shall be in accordance with MIL-B-5087 and as specified herein.

3.2.3 Mounting. The mounting base furnished with the Electronic Altimeter subsystem with the receiver-transmitter mounted shall be self locking. Basic mounting base shall be as specified in MS91405-A1B.

3.2.4 Interference Control. The design aspects of the installation program shall include such areas as preferred circuitry from the electromagnetic compatibility standpoint, RF filtering and/or by-passing required to prevent extraneous emissions and susceptibility to extraneous voltages and fields, integral suppression of interference producing devices, etc. The degree of

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shielding or isolation shall be sufficient to insure compliance with MIL-E-6051 except as specified in 3.2.12.2.c.

3.2.5 Precautions. Do not disconnect or connect any connector or remove or replace any components when power is applied to equipment.

3.2.6 Input Requirements - Electrical. The Electronic Altimeter has been designed to provide reliable performance when supplied with electrical input power with characteristics as defined in MIL-STD-704 and limits as specified below:

3.2.6.1 AC Power (single phase)

a.	Voltage limits	107 to 119 volts
b.	Frequency limits	380 to 420 Hz
c.	Power	100 VA plus 2.5 VA if 5V lighting is used

3.2.6.2 DC Power

a.	Voltage limits	25 to 28.5 volts
b.	Power	4.5 watts plus 1.4 watts for 28V lighting

3.2.7 Environmental Requirements. The Electronic Altimeter sub-system components are in accordance with the environmental requirements for Class 2 equipment designed for operation in both helicopters and aircraft as specified in MIL-E-5400 where indicated by "C" (compliance) in Tables IV-A (operating) and IV-B (nonoperating). The environmental limits of those components marked "NC" (non-compliance) are specified in 3.2.12. In addition, the RI Filter is in accordance with the environmental requirements for Class 3 equipment as specified in MIL-E-5400. Components of the Electronic Altimeter sub-system shall be installed within the limits specified.

3.2.8 Equipment Heat Dissipation. Heat dissipation data for each unit of equipment is given in Table I. The total heat dissipation of the equipment is 100 watts.

3.2.9 Total Weight. The total weight of the equipment exclusive of cables is 19.4 pounds. The weight and center of gravity of the individual units is as specified in Table I and II and Figures 2, 3 and 4 respectively.

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TABLE IV - A

Equipment Designation	Part Number or Nomenclature	OPERATING ENVIRONMENTAL SERVICE CONDITIONS												
		I	II	III	IV	V		VI		VII	VIII	IX	X	XI
						Vibration		Shock						
		Temperature	Altitude	Temperature Altitude Combined	Humidity	Normally Mounted	With Isolators Removed	Equipment	Mounting Base	Sand and Dust	Fungus	Salt Atmosphere	Explosion Proof	Interference
RT-804A/APN-171(V)	Receiver-Transmitter	NC	NC	NC	C	C	C	C	C	C	C	C	C	NC
ID-1345A/APN-171(V)	Height Indicator	NC	NC	NC	C	NC	NA	C	C	C	C	C	C	C
AS-1858/APN-171(V)	Antenna	C	C	C	C	C	NA	C	C	C	C	C	C	C
MT-4167/APN-171(V)	Mounting Base	C	C	C	C	C	C	C	C	C	C	C	C	C
F-1263/APN-171A(V)1	Radio Interference Filter	C	C	C	C	C	NA	C	C	C	C	C	C	C

NOTE: "C" denotes compliance and "NC" denotes noncompliance with specified requirements for Class 2 equipment designed for operation in both helicopters and aircraft as specified in MIL-E-5400.

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TABLE IV - B

Equipment Designation	Part Number or Nomenclature	NONOPERATING ENVIRONMENTAL SERVICE CONDITIONS												
		I	II	III	IV	V		VI		VII	VIII	IX	X	XI
						Vibration		Shock						
		Temperature	Altitude	Temperature Altitude Combined	Humidity	Normally Mounted	With Isolators Removed	Equipment	Mounting Base	Sand and Dust	Fungus	Salt Atmosphere	Explosion Proof	Interference
RT-804A/APN-171(V)	Receiver-Transmitter	C	C	C	C	C	C	C	C	C	C	C	C	C
ID-1345A/APN-171(V)	Height indicator	C	C	C	C	NC	NA	C	C	C	C	C	C	C
AS-1858/APN-171(V)	Antenna	C	C	C	C	C	NA	C	C	C	C	C	C	C
MT-4167/APN-171(V)	Mounting Base	C	C	C	C	C	C	C	C	C	C	C	C	C
F-1263/APN-171A(V)1	Radio Interference Filter	C	C		C	C	NA	C	C	C	C	C	C	C

NOTE: "C" denotes compliance and "NC" denotes noncompliance with specified requirements for Class 2 equipment designed for operation in both helicopters and aircraft as specified in MIL-E-5400.

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3.2.10 Local Consideration. Every consideration shall be given in the location of equipment and in the design of installation details to promote operator efficiency and ease of adjustment and replacement. These considerations shall not be compromised to facilitate aircraft production techniques. The completed installation shall comply as specified in MIL-I-8700.

3.2.11 Explosion Proof. This equipment is explosion-proof as defined in MIL-E-5400.

3.2.12 Detailed Requirements

3.2.12.1 Antennas. Installation and location of airborne antennas shall be in accordance with Military Specification MIL-STD-877 and as specified herein.

a. Residual Altitude. Antennas shall be installed so that the residual altitude is between one and fifty feet.

b. Antenna Placement. A reasonably flat area on the underside of the aircraft shall be selected. There shall be no projections extending into a 140-degree cone about the center of either antenna.

c. Antenna Isolation. Antennas shall be spaced at least 24 inches and preferably 36 inches apart, center to center. This spacing will provide the needed 85 decibels of attenuation between the transmitting antenna and the receiving antenna.

d. Antenna Orientation. Antennas shall be oriented so that the connectors are perpendicular to an imaginary line drawn through the two antenna centers. This arrangement allows the antenna H-planes of radiation to couple and prevents the E-planes of radiation from coupling.

e. Antenna Mounting. Antennas may be mounted flush with or protruding from the aircraft skin. Each antenna shall be secured in place with twelve 8-32UNC-2A, 82-degree flat head machine screws, lock washers, and 8-32 hexagon nuts. Metal-to-metal contact between the antennas and the aircraft skin for electrical purposes is not necessary, but is considered preferable.

f. Limits to Normal Operation Attitude. Antennas shall be so mounted that the exposed radiating surfaces outside the aircraft skin are level with each other and not more than 10 degrees from horizontal with the aircraft in normal flight attitude.

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g. Precautions. The following precaution shall be taken: The exposed radiating surfaces of the antennas shall never be painted, lacquered, varnished or otherwise coated with any material which may attenuate the transmission or reception of radio frequency energy.

h. Mounting Instructions. Refer to Figure 2 to assist in mounting the antennas. Rectangular holes are required in the skin of the aircraft at each point that an antenna is to be installed. Observe the installation limitations as specified in 3. 2. 12. 1. a through 3. 2. 12. 1. g and mount the antennas in the selected locations.

3. 2. 12. 2 Receiver-Transmitter and Mounting Base

a. Environmental Conditions. This equipment shall operate satisfactorily under any of the environmental service conditions of MIL-E-5400 for Class 2 equipment except for the following operating limits:

Temperature (degrees F)	+ 160 max
Altitude	Up to 50, 000 ft.

Vibration - In addition to the requirements of MIL-E-5400 the receiver-transmitter mounted on the mounting base shall operate satisfactorily when subjected to the following vibration level:

± 0. 050 inch amplitude (0. 8G) at 13 cps - 50 hrs. per axis

b. Interference Control. The generation of radio frequency interference by the equipment and the vulnerability of the equipment to radio frequency interference shall be controlled within the limits of MIL-I-6181 except as follows:

Receiver Port Radiation. Since the local oscillator frequency is the same as the transmitter frequency, compliance with the local oscillator radiation requirements of MIL-I-6181 shall not be required for the frequency band between 4290 and 4310 MHz.

c. Location of Unit

1) Environment. The receiver-transmitter shall not be installed in a location where operation will result beyond the limits as specified in 3. 2. 12. 2. a.

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2) Cable Lengths: The receiver-transmitter and mounting base shall be located with respect to the selected location of the Antennas as specified in 3.2.12.1 so that a total coaxial cable length of 20 feet or less may be used to interconnect the Electronic Altimeter components. If it is necessary to have a total coaxial cable length greater than 20 feet, the equipment manufacturer should be notified so that suitable recommendations can be made regarding the type of cable to use and new procedures that may be necessary.

3) Airflow and Access. The receiver-transmitter location shall be sufficiently clear of surrounding objects to allow adequate airflow through the unit and to allow removal of the cable connections.

4) Sway Clearance. Refer to Figure 3 for sway clearance and receiver-transmitter dimensions when selecting a location so that adequate spacing between the unit and nearby objects may be obtained.

d. Limits to Normal Operation Attitude. The receiver-transmitter may be operated in any attitude. However, to allow the shock absorbing capabilities of the base to function properly, the unit should be mounted so that it is level under normal flight conditions.

e. Electrical Bonding. Special care shall be taken to insure a good electrical bond between the base bonding straps and the mounting surface structure. The bonding straps shall be positioned such that the sway motion of the base is not restricted.

f. Mounting Instructions.

1) Refer to Figure 3 for information on mounting the receiver-transmitter and base. Observe the installation limitation as specified in 3.2.12.2.d and permanently attach base to the aircraft structure using sixteen sets of 3/16-inch mounting hardware.

2) Mount the receiver-transmitter on the base engaging the two guide pins on the base into two holes on the rear panel of the receiver-transmitter. Secure the base-mounted fasteners over the two brackets on the lower front corners of the receiver-transmitter.

3.2.12.3 Height Indicator

a. Environmental Conditions. This equipment shall operate satisfactorily under any of the environmental service conditions of MIL-E-5400 for Class 2 equipment except for the following operating limits:

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Temperature (degrees F)

Altitude Up to 50,000 ft.

Vibration Hard Mounted: Curve III of MIL-E-5400

b. Location of Unit

1) Environment. The height indicator shall not be installed in a location where operation will result beyond the limits as specified in 3.2.12.3.a.

2) Operator Considerations. The height indicator is designed to be mounted in the aircraft instrument panel. The location on the panel should be selected for convenience of the operator in viewing the scale and operating the control switch. Space should be available at the rear of the unit to allow for removal of the leadwire cable connector.

c. Electrical Bonding. Special care shall be taken to insure a good electrical bond between the indicator mounting flange and the instrument panel.

d. Mounting Instructions. Refer to Figure 4 for information on mounting the height indicator. Clearance holes shall be provided in the aircraft instrument panel for the body of the height indicator and the four mounting screws. Insert the height indicator in the indicator clearance hole from the front side of the panel and secure in place with four sets of 5/23-inch mounting hardware. Connect the leadwire cable connector to the height indicator from the rear of the panel.

3.2.12.4 Radio Interference Filter.

a. Location of Unit. The radio interference filter shall be located with respect to the selected locations of the transmitting antenna and receiver-transmitter so that a total coaxial cable length of 20 feet or less may be used to interconnect the components as specified in 3.2.12.2.d.

b. Mounting Instructions. Refer to Figure 5 for information on mounting the radio interference filter. Observe the installation limitations as specified in 3.2.12.4.a and mount the radio interference filter to the aircraft structure using four sets of 8-32 mounting hardware. Use standoff spacers to provide clearance between the cable connectors and adjacent surfaces if necessary.

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3.2.12.5 Cabling. Refer to Figure 6, AN/APN-171A(V)1 Electronic Altimeter Set External Wiring, and Figure 7, AN/APN-171A(V)1 Electronic Altimeter Set, Cabling Diagram, as attached.

a. Location of Unit. The radio interference filter shall be located with respect to the selected locations of the transmitting antenna and receiver-transmitter so that a total coaxial cable length of 20 feet or less may be used to interconnect the components as specified in 3.2.12.2.d.

b. Mounting Instructions. Refer to Figure 5 for information on mounting the radio interference filter. Observe the installation limitations as specified in 3.2.12.4.a and mount the radio interference filter to the aircraft structure using four sets of 8-32 mounting hardware. Use standoff spacers to provide clearance between the cable connectors and adjacent surfaces if necessary.

3.2.12.5 Cabling. Refer to Figure 6, AN/APN-171A(V)1 Electronic Altimeter Set External Wiring, and Figure 7, AN/APN-171A(V)1 Electronic Altimeter Set, Cabling Diagram, as attached.

a. Cabling Instructions.

1) Cable lengths and plug requirements are determined by the particular installation. Cables and MS type connectors are not supplied with the equipment. Cables and connectors shall be installed in accordance with MIL-I-8700.

2) Extreme care shall be used in constructing the coaxial cables used to interconnect the receiver-transmitter, radio interference filter, and the antennas to prevent a high voltage standing wave ratio. Each coaxial cable shall be checked as specified in 3.2.12.5.c prior to installation in the aircraft.

3) Construct the leadwire cabling according to Figure 6. The leadwire cable may be broken once for the installation of a terminal strip at any convenient and accessible location in the aircraft. The terminal strip shall form the junction for the leadwire cable to the height indicator and the leadwire cable to the receiver-transmitter and connections to the power sources and other aircraft systems as required.

4) Refer to Figure 6 for indicator lighting connector pin assignments. Note that either 28 volt dc or 5 volt (ac or dc) may be used. The preferred lighting voltage is 5 volts where a choice exists.

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5) The indicator provides a means for dimming the low altitude warning lamp. Refer to Figure 6 for connector pin assignments. If dimming is not required, the appropriate connector pin shall be grounded. If dimming is required, a dimming rheostat, supplied by the installing activity, should be located in a position convenient to the operator.

b. Precautions

1) Avoid sharp bends.

2) Provide sufficient length for disconnecting cables from units.

3) Total coaxial cable length shall not exceed 20 feet as specified in 3.2.12.2.d2).

4) Avoid any excessive loading in any direction of mounting base resulting from tension or weight of cables by providing slack or support for the cables as required.

5) Do not route any cables beneath the frame of the mounting base.

c. Voltage Standing Wave Ratio (VSWR) and Continuity Check. Before connecting the cables to the major units, check the leadwire and coaxial cables as follows:

1) Check the interconnecting coaxial cables for VSWR, using an rf generator, standing wave bridge, indicator, and a suitable 50-ohm resistive termination. The rf generator shall be able to provide an output frequency of 4,300 MHz. The VSWR at 4,300 MHz shall be 1.4 to 1.0 or less.

2) Check the interconnecting leadwire cabling for continuity using a multimeter. Continuity shall exist in accordance with Figure 6.

3.2.13 Initial Zero Altitude Calibration. The receiver-transmitter must be calibrated once after installation in an aircraft to compensate for variations in RF delays for each installation.

3.2.13.1 Perform initial calibration as follows:

a. Remove the receiver-transmitter from the base and remove the dust cover from the receiver-transmitter.

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b. Turn the control knob on the height indicator clockwise from the OFF position. Allow five minutes for the receiver-transmitter to reach operating temperature. The flag should disappear within two minutes indicating that the Electronic Altimeter Set is tracking the ground.

c. Observe the indicated altitude on the height indicator. No adjustment is necessary if the indicated altitude is exactly zero. If the indicated altitude is not exactly zero, proceed to d. below.

d. Adjust A2R31 as shown in figure 8 for an altitude indication of exactly zero feet.

e. Turn the height indicator control know counterclockwise to OFF.

f. Replace the dust cover on the receiver-transmitter and install the receiver-transmitter on the base.

NOTE: If the antenna height from the ground with the aircraft in the parked position is different from the antenna height in the landing position, the above adjustment should be made to a value offset from zero by the difference in these two antenna heights so that the Altimeter Set will indicate zero feet in the landing position.

3.3 Equipment Performance Requirements. The applicable tests in Section 4 shall be conducted on the receiver-transmitter, height indicator, and radio interference filter of the Electronic Altimeter sub-system to insure compliance with the following requirements. Compliance with the specified equipment performance requirements shall be verified prior to installing the sub-system in an aircraft. Performance testing is not required for the antennas or mounting base.

3.3.1 Test Setup. Equipment performance tests require the use of the UG2308AA01 Radar Altimeter System Tester. A permanently located test setup shall be provided to include the Radar Altimeter System Tester and other test equipment listed in Table III or equivalent. The arrangement shall provide ready access to the components parts and adjustments. The input power sources shall be capable of providing power within the following limits:

AC Power:	115 ± 1v
	400 ± 20 Hz
DC Power:	27.5 ± 0.5 v

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In the procedures that follow, mention of the tester panel components will be in capital letters as they appear on the tester panel to aid in their identification and location.

3.3.1.1 Zero Calibration Cable. The receiver-transmitter zero calibration procedures and the receiver-transmitter test procedures require a zero calibration cable. This cable simulates the travel time of an RF pulse from the transmitter to the ground and back to the receiver when the aircraft is on the ground. This travel time shall be simulated during bench testing to allow for proper adjustment of the receiver-transmitter for zero altitude conditions. Fabricate the zero calibration cable as instructed in 4.5.2.

3.3.2 Equipment Condition (See 4.5.3). The equipment shall be free of dust, dirt, foreign or packing material and any external or internal damage. All external controls shall operate freely and properly.

3.3.3 Tests (See 4.5.4). Performance testing of the receiver-transmitter and height indicator shall consist of the following tests:

Receiver-Transmitter Tests

a. Initial operation

b. Power Supply Voltages

c. Zero altitude tests

   1) Track indication

   2) Zero altitude acquisition

   3) Altitude track sensitivity

   4) Zero altitude acquisition sensitivity

   5) Push-to-test operation

d. Altitude accuracy test

e. Far range push-to-test operation

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Height Indicator Tests

a. Scale accuracy test

b. Loss of track test

c. Low level warning test

d. Indicator response time

Performance requirements are given in the following paragraphs.

3.3.3.1 Initial Operation (See 4.5.5). When power is applied to the equipment, air movement at the exhaust openings on the front of the receiver-transmitter panel and a smooth sound from the blower shall be noted. With the MONITOR switch in the -50V position, a reading of -50 ±3 VDC will be observed within two minutes.

3.3.3.2 Power Supply Voltages (See 4.5.6). The power supply voltages shall be within the limits shown in Table V.

TABLE V. Power Supply Voltages

Test Jack No.	Switch Position	Volts Minimum	Volts Maximum
3	120V	+114	+126
4	80V	+79	+81
5	28V	+24	+32
18	15V	+14.85	+15.15
11	-6.3V	-6.23	-6.37
19	-15V	-14.85	-15.15
10	-50V	-49.875	-50.125

3.3.3.3 Track Indication (See 4.5.7). The test set TRACK light shall be on.

3.3.3.4 Zero Altitude Calibration (See 4.5.8). The external range voltage at SYSTEM tip jacks E and T (ground) shall be zero ±0.007 vdc.

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3.3.3.5 Zero Altitude Acquisition (See 4.5.9). With the MODE switch at FORCED SRCH, the TRACK light shall go out and the external range voltage shall be -46.7 ±1.0 vdc. With the MODE switch returned to NORM, the track light shall come on.

3.3.3.6 Altitude Track Sensitivity (See 4.5.10). The receiver-transmitter shall lose track at 76 ±3 db.

3.3.3.7 Zero Altitude Acquisition Sensitivity (See 4.5.11). The receiver-transmitter shall regain track at 73 ±4 db.

3.3.3.8 Push-to-Test Operation (See 4.5.12). The external range voltage shall be -0.80 ±0.12 vdc and the indicator reading shall be 100 ±15 ft.

3.3.3.9 Altitude Accuracy Test (See 4.5.13). The external range voltage shall be within limits shown in Table VI for the specified simulated range and attenuation.

TABLE VI. Altitude Accuracy Limits

Simulated Altitude (Feet)	Total Loop Attenuation (DB)	External Range Voltage
0	52 ±2	0 ±0.012
250	88 ±2	-2.0 ±0.032
500	88 ±2	-4.0 ±0.050
1000	88 ±2	-8.0 ±0.092
2500	88 ±2	-20.0 ±0.212
5000	88 ±2	-40.0 ±0.412

3.3.3.10 Far Range Push-to-Test Operation (See 4.5.14). External range voltage shall be -0.800 ±0.120 vdc.

3.3.3.11 Scale Accuracy Test (See 4.5.15). The applied altitude voltage shall be within limits and tolerance shown in Table VII for the indicated altitude.

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TABLE VII. Indicator Scale Accuracy Tests

Step	Adjust Altitude Indication (Feet)	Applied Voltage Limits (Volts DC)	Repeatability Voltage Tolerance (Volts DC)
1	0	0.000 to 0.012	±0.010
2	150	-1.194 to -1.213	±0.010
3	500	-3.920 to -4.080	±0.020
4	3000	-23.528 to -24.471	±0.140

3.3.3.12 Loss of Track Test (See 4.5.16). The indicator pointer shall be behind the NO TRACK mask and the flag shall shift to its no-track position for the specified input conditions.

3.3.3.13 Low Level Warning Test (See 4.5.17). The LOW warning lamp on the indicator and the LOW ALT WARNING LAMP on the tester shall operate within the limits of Table VIII.

TABLE VIII. Low Level Warning Voltage Limits

Step	Index Setting (Feet)	Applied Voltage Limits When	
		Lamp Lights	Lamp Goes Out
1	50	-0.368 to -0.432	-0.400 to -0.480
2	150	-1.168 to -1.232	-1.200 to -1.280
3	600	-4.640 to -4.960	-4.800 to -5.176
4	3000	-23.144 to -24.860	-24.000 to -26.136

3.3.3.14 Response Time (See 4.5.18). The indicator response time shall not exceed two seconds.

3.4 Engineering Approval Requirements. Engineering approval is required prior to final acceptance by the procuring activity of the Electronic Altimeter sub-system installation and operation. The applicable tests specified in Section 4 shall be conducted on a prototype or first production model aircraft

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of a given contract and on the first aircraft following a major change in either the aircraft or the equipment installation design to insure compliance with the engineering approval requirements specified herein. The aircraft selected shall be fully configured (electronics, armament, armor plating, etc.) as specified in the aircraft contract unless otherwise approved by the procuring activity. If the installation is part of a retrofit program, all applicable MWO's shall be applied prior to conducting the engineering approval tests. Engineering approval requirements shall consist of preflight, flight and EMC requirements as specified in 3.4.1 through 3.4.3.

3.4.1 Preflight. Compliance with the following preflight requirements shall be verified prior to determining compliance with the flight requirements as specified in 3.4.2 and after verification of equipment performance requirements as specified in 3.3 to insure the equipment installation and operation is in accordance with the following preflight requirements. Engineering preflight tests (see 4.6.1) shall be performed with Electronic Altimeter sub-system installed in the aircraft in accordance with 3.2.

3.4.1.1 Power Sources (See 4.6.1.1). Electrical power shall be obtained from the aircraft power busses. The supply voltage measured at the input terminals of the Electronic Altimeter shall be as follows:

AC Power: 107 to 119V

DC Power: 25 to 28.5V

3.4.1.2 Initial Operation (See 4.6.1.2). The OFF flag shall disappear from view approximately three minutes after the equipment is turned on.

3.4.1.2 Indicator Illumination (See 4.6.1.3). The indicator shall be illuminated.

3.4.1.4 Warning Lamp and Push-to-Test Operation (See 4.6.1.4). The indicator shall display an altitude of 100 $\pm$15 feet and the LOW warning lamp shall go off when push-to-test is actuated. The LOW warning lamp shall light when the push-to-test knob is released.

3.4.2 Flight. Compliance with the following flight requirements shall be verified after compliance with the preflight requirements specified in 3.4.1 has been verified. Engineering flight tests (see 4.6.2) shall be performed to insure the Electronic Altimeter sub-system performance is in accordance with the following flight requirements while all the "normal" in-flight equipment in the aircraft are functioning.

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3.4.2.1 Comparison with Standard (See 4.6.2.1). Accuracy of indication shall be within the limits of Table IX.

TABLE IX. Flight Test Accuracy Limits

Standard (Feet)	Height Indicator (Feet)
50	50.0 ± 8.1
100	100.0 ± 9.6
190	190.0 ± 12.4
1,000	1,000.0 ± 45.0
5,000	5,000.0 ± 232.0

3.4.2.2 Antenna Beamwidth (See 4.6.2.2). At aircraft roll angles up to 30 degrees the height indicator shall continue to indicate 1000 ±45 feet.

3.4.2.3 Low Level Warning Operation (See 4.6.2.3). On descent the LOW warning lamp shall come on as the pointer passes directly adjacent to the index. On ascent the lamp will go out within 55 feet above the index.

3.4.2.4 Maximum Altitude Acquisition (See 4.6.2.4). When the aircraft is above the maximum altitude range of the equipment, the indicator pointer shall remain behind the NO TRACK mask and the OFF flag shall be visible. The pointer shall begin to move from behind the mask and the OFF flag shall disappear from view before the aircraft descends below 5000 feet indicating that a tracking condition has been established.

3.4.2.5 Push-to-Test Operation (See 4.6.2.5). The indicated altitude shall be 100 ±15 feet.

3.4.2.6 Zero Reference (See 4.6.2.6). The height indicator reading at touchdown shall be zero ±5 feet.

3.4.3 Electromagnetic Compatibility (See 4.3). Compliance with the complete functional electrical-electronic requirements specified in the USAECOM approved, contractor prepared, EMC control and EMC test plan required in 3.1.11 shall be verified.

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3.5 Production Approval Requirements. Production approval is required prior to final acceptance by the procuring activity of the Electronic Altimeter sub-system installation and operation. The applicable tests specified in Section 4 shall be conducted on every aircraft submitted for acceptance to insure compliance with the production approval requirements as specified herein. The aircraft shall be fully configured (electronics, armament, armor plating, etc.) as specified in the aircraft contract unless otherwise approved by the procuring activity. If the installation is part of a retrofit program, all applicable MWO's shall be applied prior to conducting the production approval tests. Production approval requirements shall consist of preflight, flight and EMC requirements as specified in 3.5.1 through 3.5.3.

3.5.1 Preflight. Compliance with the following preflight requirements shall be verified prior to determining compliance with the flight requirements as specified in 3.5.3 and after verification of equipment performance requirements as specified in 3.3. Production preflight tests shall be performed with the Electronic Altimeter sub-system installed in the aircraft in accordance with 3.2 to insure the equipment installation and operation is in accordance with the following preflight requirements.

3.5.1.1 General. Production preflight requirements shall be verified on every aircraft submitted for acceptance and shall consist of those requirements as specified in 3.4.1.2, 3.4.1.3 and 3.4.1.4.

3.5.2 Flight. Compliance with the following flight requirements shall be verified after compliance with the preflight requirements specified in 3.5.1 has been verified. Production flight tests shall be performed to insure the Electronic Altimeter sub-system performance is in accordance with the following flight requirements while all "normal" in-flight equipments in the aircraft are functioning.

3.5.2.1 General. Production flight requirements shall be verified on every aircraft submitted for acceptance and shall consist of those requirements as specified in 3.4.2.2, 3.4.2.3, 3.4.2.4, 3.4.2.5 and 3.4.2.6. Prior to performance of any tests, and before the equipment is airborne, the control knob on the height indicator shall be rotated clockwise from the OFF position and the equipment allowed five minutes to stabilize.

3.5.3 Electromagnetic Compatibility (See 4.3). Compliance with the limited functional electrical-electronic requirements specified in the USAECOM approved, contractor prepared, EMC control and EMC test plan required in 3.1.11 shall be verified.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Examinations and Tests. The examinations and tests required to assure conformance with the requirements of Section 3 are classified as follows:

a. Electromagnetic compatibility tests

b. Installation examination

c. Equipment performance tests

d. Engineering approval and Production approval tests

4.3 Electromagnetic Compatibility Tests. Under the test conditions and procedures specified in the USAECOM approved, contractor prepared, EMC Control and EMC test plans, demonstrate compliance with 3.4.3 and 3.5.3.

4.4 Installation Examinations. All Electronic Altimeter sub-system electronic equipment, whether Government furnished or contractor furnished, which is installed by the contractor shall be examined to determine compliance with 3.2.

4.5 Equipment Performance Tests.

4.5.1 Input Power. All power sources shall be in compliance with 3.3.1.

4.5.2 Zero Calibration Cable. Fabricate a zero calibration cable (See 3.3.1.1) as instructed below. Round off all calculations and measurements to the nearest inch.

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4.5.2.1 Obtain the total length of the coaxial cables interconnecting the receiver-transmitter, RI filter, and antennas in the aircraft. Add 116 inches to simulate the delay of the RI filter.

4.5.2.2 Obtain the height above ground of each of the two antennas. Compute the average of the two heights.

4.5.2.3 Calculate the required zero cable length by multiplying the average antenna height found in 4.5.2.2 by 1.34 and add this figure to the cable length obtained in 4.5.2.1.

4.5.2.4 Subtract 2 1/2 feet representing the delay of the test set attenuator panel circuit and 13 feet for the cable assemblies (PN 1995994 & 1995996, part of UG2308AA01). This result will be the zero cable length.

4.5.2.5 Use RG214/U coaxial cable as specified in MIL-C-17/75 (or equivalent) and TNC plug connectors as specified in MIL-C-23329/4 to fabricate a cable with a length (including connectors) equivalent to the length obtained in 4.5.2.4.

4.5.2.6 Check the coaxial cable for voltage standing wave ratio (VSWR) at 4,300 MHz (See 3.2.12.5c). The VSWR shall be 1.2 to 1.0 or less.

NOTE: If the installation is changed to require different coaxial cable lengths or different antenna heights above ground, a new zero calibration cable shall be fabricated to match the characteristics of the new installation.

4.5.3 Visual Inspection. All equipment shall be inspected for external and internal damage. Remove all dust, dirt, foreign or packing material. Check all external controls to see that they operate freely and properly. Remove equipment covers and examine for foreign material and loose or damaged parts for compliance with 3.3.2. Clean, tighten, repair and replace in accordance with 3.1.6.

4.5.4 Preparation. Remove the test cover from the receiver-transmitter and connect the receiver-transmitter, height indicator and radar altimeter system tester as shown in Figure 9. Use the cables provided with the tester and the zero calibration cable fabricated as instructed in 4.5.2. Set the tester switches to the initial positions shown in Table X. When determining the series attenuation values specified during testing, include the recorded attenuation value of the calibration cables, the variable and fixed attenuators, and the recorded attenuation of the delay cable, if it is being used, that is provided

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in the calibration chart supplied with the tester. Unless otherwise specified, the series attenuation used in any particular test remains the same as in the preceding test. When a test specifies a value of attenuation, observe the following:

TABLE X. Initial Position of Tester

Panel	Control	Initial Position
Control and Simulation	Tester Power	ON
	Mode Control	OFF
	Fixed Wing/Rotary Wing Mode	FIXED WING
	Manual Slew	NORM
	Auto Plt Rdr Altm Engage	OFF
	Preset Alt	OFF
	Rate Test	OFF
	Monitor	-50V (X20)
Range and Rate Simulation	Range/Rate Select	RANGE
	Range Control	0
	RT-805-RT-804/RT-829	RT-804/RT-829
	Inbound/Outbound	HOLD
	Video	POS
	PWC	AUTO
Attenuator	40-90DB/85-135DB	40-90 DB

4.5.4.1 When connecting or changing attenuator cables to the receiver-transmitter, turn off the power input.

CAUTION: Before applying power to the receiver-transmitter, make certain that the transmitter rf connector J2 is either connected according to Figure 9 or is terminated by a 50-ohm resistive load (J7 of the attenuator panel of the UG2308AA01 Tester provides 50-ohm load). Failure to do so may result in damage to the transmitter module in the receiver-transmitter.

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4.5.4.2 The attenuator panel contains a 20 db fixed attenuator in series with a zero-to-50 db variable attenuator. For attenuation values of 40 to 90 db, the number 1995994 transmitter cable, which has a fixed 20 db attenuator, is used. Set the ATTENUATOR panel switch to 40-90 db. Connect the attenuation loop input to J1, and output to J8 on the attenuator panel.

4.5.4.3 For attenuation values of 85 to 135 db, move the attenuation loop output from J8 to J6, and set the ATTENUATOR panel switch to 85-135 db. This connects the 45.7 db delay cable (representing about 125 radar feet) in series with the attenuator loop.

4.5.4.4 Receptacle J7 on the attenuator panel connects to a 50-ohm termination. This must be used at all times whenever the transmitter is energized, except when using the attenuation loop for measurements, or when the transmitter test connector is removed.

4.5.4.5 Receptacles J2 and J3 connect to the wavemeter and crystal detector unit used in checking transmitter frequency. Receptacles J4 and J5 connect to a second crystal detector used in power level checks. Further details regarding testing of the Electronic Altimeter Set or operation of the UG2308AA01 Radar Altimeter System Tester may be found in Technical Manuals NAVAIR 16-30-APN171-1 or NAVAIR 16-45-1205-1 respectively.

4.5.5 Initial Operation. With the MODE CONTROL switch set to XCVR/MODULE, check for appreciable air movement at the blower opening on the receiver-transmitter panel. A smooth sound shall be noted if the blower is operating properly. (Normal air flow is about 32 cubic feet per minute.) Observe the MONITOR meter. A -50 $\pm$3 vdc reading must be otained in two minutes. If not, turn off the tester power. Remove the power supply module from the receiver-transmitter. Install power supply known to be operational before resuming tests.

4.5.6 Power Supply Voltages. Set the MONITOR switch to each of its seven positions from +120 V to -50V. Voltage reading on the MONITOR voltmeter shall be approximately the same as listed in Table V if the power supply is functioning. For exact reading, use the differential ac-dc voltmeter and check voltage readings according to Table V, using TEST tip jack 13 for ground.

4.5.7 Track Indication. Check for indication that the receiver-transmitter acquires track. With the equipment connected as shown in Figure 9, set the total attenuation to 52 db. The tester TRACK light shall be on. Use the multimeter to measure voltage between SYSTEM tip jacks C and B (ground), and TEST tip jacks 17 and 14. Reading shall be 28 $\pm$4 vdc.

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NOTE: It must be determined that the zero cable used is simulating the correct zero altitude for the particular aircraft installation.

4.5.8 Zero Altitude Calibration. Operate the receiver-transmitter with the shorting plug in place on the transmitter test connector J4.

4.5.8.1 Allow five minutes for the receiver-transmitter to reach operating temperature. The TRACK light should come on after approximately 2.5 minutes.

4.5.8.2 Connect the differential voltmeter between the SYSTEM tip jacks E and T (ground). If the voltage is zero ± 0.007 vdc, no adjustment is necessary. If the voltage is outside these limits, proceed to 4.5.8.3.

4.5.8.3 Adjust variable resistor A2R31 (see Figure 8) for a voltage indication of zero ± 0.007 vdc.

4.5.9 Zero Altitude Acquisition. Check reacquisition of target at zero altitude, after breaking track.

4.5.9.1 With the total attenuation set to 52 db, set the MODE switch to FORCED SRCH. The TRACK light shall go out.

4.5.9.2 Use the multimeter to check for voltages at the following TEST and SYSTEM tip jacks:

a. Zero ± 0.5 vdc between tip jacks C and B (ground).

b. Negative (-) 46.7 ± 1.0 vdc between tip jacks E and T (ground).

c. Zero ± 0.5 vdc between tip jacks 17 and 14 (ground).

4.5.9.3 With the multimeter remaining connected to tip jacks 17 and 14, set the MODE switch to NORM. The TRACK light shall come on, the voltage reading shall be 24 to 32 vdc, indicating track acquisition.

4.5.10 Altitude Track Sensitivity. Check the tracking sensitivity by measuring the attenuation setting required to cause loss of track.

4.5.10.1 With the total attenuation remaining at 52 db and MODE switch set to NORM, connect the multimeter to TEST tip jacks 12(+) and 14(-). The voltage observed (internal range) shall be constant at nearly zero vdc.

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4.5.10.2 Slowly increase the attenuation setting until a sudden voltage change toward positive is noted, and the TRACK light goes out. The voltage shall vary from zero to 70 vdc at about 2 Hz. Total attenuation required shall be 76 ± 3 db.

4.5.11 Zero Altitude Acquisition Sensitivity. Check the acquisition sensitivity by measuring the attenuation level at which track is reacquired.

4.5.11.1 With the multimeter remaining connected to tip jacks 12(+) and 14(-), the MODE switch at NORM, set the total attenuation to about 90 db.

4.5.11.2 Slowly decrease the attenuation. A point should be observed where the internal range voltage stops sweeping, stops at nearly zero vdc, shifts toward negative, and the TRACK light comes on.

4.5.11.3 Total attenuation shall be 73 ± 4 db. Record the actual value obtained.

4.5.12 Push-To-Test Operation. Check the external range voltage and height indicator reading when the push-to-test circuit is activated.

4.5.12.1 Connect the differential ac-dc voltmeter to SYSTEM tip jacks E(-) and T(+). Set the total loop attenuation at 69 to 75 db. Set the MODE CONTROL switch to IND & XCVR.

4.5.12.2 Press the tester PUSH TO TEST switch. The external range voltage indicated on the voltmeter shall be -0.80 ± 0.12 vdc. The indicator readings shall be 100 ± 15 feet.

4.5.13 Altitude Accuracy Test. Check the accuracy of the external range voltage for applied steps of simulated altitude.

4.5.13.1 With the tester RANGE CONTROL set to zero, connect the differential ac-dc voltmeter to SYSTEM tip jacks E(-) and T(+).

4.5.13.2 Remove the shorting plug from the transmitter receptacle J4. Connect the calibration cable from the transmitter receptacle J4 to the CAL receptacle J5 on the tester RANGE AND RATE SIMULATION panel. Adjust the total loop attenuation to 52 ± 2 db.

4.5.13.3 Set the PWC switch to NARROW. Adjust the T-ZERO DELAY control to obtain a zero altitude reading. Press and release the PUSH TO TEST switch.

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4.5.13.4 Turn off power to the receiver-transmitter by setting the tester MODE CONTROL switch to OFF, or by turning off the indicator power switch. After two or three minutes, the zero altitude voltage reading at tip jack E shall return to within +5 mv of the same value obtained in 4.5.13.3.

4.5.13.5 Check the altitude accuracy of the receiver-transmitter for the simulated altitudes and loop attenuations of Table VI.

4.5.14 Far Range Push-To Test. With the same test setup as in the external range voltage accuracy and tracking sensitivity tests, perform the push-to-test check at medium and at high altitude conditions.

4.5.14.1 With the MODE switch at NORM and PWC AUTO/NARROW switch at NARROW, set the total loop attenuation to 100 $\pm$ 2 db.

4.5.14.2 Set the RANGE CONTROL selector to 1000. Press the PUSH to TEST switch. The external altitude voltage shall drop to -0.80 $\pm$0.12 millivolts and the TRACK light shall be on. Release the PUSH TO TEST switch.

4.5.14.3 Set the RANGE CONTROL selector to 4000. Set the PWC switch to AUTO.

4.5.14.4 Press the PUSH TO TEST switch. Results shall be the same. Release the PUSH TO TEST.

4.5.15 Scale Accuracy Test. Conduct scale accuracy tests as follows:

4.5.15.1 Set the MODE CONTROL switch to IND.

4.5.15.2 The indicator internal lighting and the tester POWER ON indicator light shall be on. The OFF flag in the window on the face of the indicator shall disappear from view.

4.5.15.3 Rotate the control knob on the height indicator clockwise from the OFF position. The AUX IND light in the test set INDICATOR POWER section shall be on.

4.5.15.4 In the steps that follow, use the differential ac-dc voltmeter and monitor the voltage at the ALT SIG and GRD tip jacks on the tester INDICATOR CONTROLS section.

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a. Perform a scale accuracy test of the height indicator according to Table VII. Use the ALT SIGNAL control to apply a voltage representing an altitude signal to the indicator under test. Adjust the control until the altitude indicated by the indicator is exactly as specified in Table VII.

b. Measure the applied voltage with the vtvm. The voltage value shall be between the two values listed in the table for the indicated altitude.

c. When a repeatability voltage is specified, record the exact voltage for the indicated altitude. Adjust the voltage to move the indicator pointer either up or down scale. Readjust the voltage to return the indicator pointer to the indicated altitude. Measure the voltage with the vtvm. The voltage shall be within the repeatability voltage tolerance listed in the table.

4.5.16 Loss of Track Test. Conduct loss of track tests as follows:

4.5.16.1 Adjust the applied voltage for a vtvm indication of -46.7 $\pm$ 0.7 vdc.

4.5.16.2 The indicator pointer tip shall be completely behind the NO TRACK mask.

4.5.16.3 Adjust the voltage to return the pointer to zero. Press the tester NO TRACK switch. The indicator flag should shift to its no-track position.

4.5.17 Low Level Warning Test. Conduct low level warning test as follows:

4.5.17.1 Rotate the indicator control knob to move the low level index to the maximum range end of the indicator scale. The index shall not bind or touch any part of the dial face or pointer during travel.

4.5.17.2 Rotate the control knob to its maximum counter-clockwise position, removing power from the indicator.

4.5.17.3 Adjust the applied voltage to -5.0 volts as indicated on the vacuum tube voltmeter. The indicator pointer shall remain at zero.

4.5.17.4 Rotate the control knob clockwise from the OFF POSITION to restore power to the indicator. The indicator pointer shall move away from the zero point. Adjust the applied voltage to return the pointer to zero.

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4.5.17.5 Test the low level warning index according to Table VIII. Set the low level index to the desired altitude. Adjust the applied voltage to drive the indicator pointer down scale until the LOW warning lamp lights. The applied voltage shall be within the limits specified in the table. Repeat for each altitude given in Table VIII.

4.5.17.6 Repeat 4.5.17.5 except drive the indicator pointer upscale until the LOW warning lamp goes out. The lamp shall go out within the limits specified in Table VIII.

4.5.17.7 Adjust the applied voltage to return the pointer to zero.

4.5.18 Response Time. Conduct indicator response time tests as follows:

4.5.18.1 Rotate the indicator control knob to the OFF position to remove power from the indicator.

4.5.18.2 Adjust the ALT SIGNAL control to provide -43 vdc as measured by vtvm.

4.5.18.3 Restore power to the indicator with the indicator control knob and start a stop watch at the same time. Measure the time necessary for the indicator to reach maximum altitude. The response time shall not exceed two seconds.

4.6 Engineering Approval and Production Approval Tests

4.6.1 Preflight Tests

4.6.1.1 Electrical Power Sources. With all electrical equipment which is connected to the same busses as the Electronic Altimeter Set turned "ON" and operating in the normal manner, turn the Electronic Altimeter Set "ON" with the indicator control knob. Under this condition measure the ac voltage level at the indicator connector (2P1 of Figure 6) pin A to pin B and the dc voltage level at the receiver-transmitter power connector (1P1 of Figure 6) pin c (+) to pin B with the differential voltmeter to determine compliance with 3.4.1.1.

4.6.1.2 Initial Operation. Turn the control knob on the height indicator clockwise from the OFF position. Allow five minutes for the receiver-transmitter to reach operating temperature. The OFF flag shall disappear from view within approximately three minutes.

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4.6.1.3 Indicator Illumination. Observe that the height indicator is illuminated by the lamps in the mask on the front of the indicator housing.

NOTE: The appropriate aircraft lighting bus must be energized.

4.6.1.4 Warning Lamp and Push-To-Test Operation.

a. Turn the control on the height indicator clockwise until the low-level index is positioned at any altitude graduation between 10 and 80 feet on the indicator scale. The LOW warning lamp shall light. If a dimming control is provided for the low altitude warning lamp (See 3.2.12.5a5), verify that the intensity of the LOW warning lamp is adjustable.

b. Press and hold the control knob to actuate the push-to-test condition.

c. The pointer on the indicator shall indicate an altitude of 100 ± 15 feet to indicate proper operation of the system. The LOW warning lamp shall be off.

d. Release the control knob to remove the push-to-test condition. The indicator pointer shall return to zero. The LOW warning lamp shall light.

e. Rotate the control knob to position the low-level index to zero.

4.6.2 Flight Tests. Prior to performance of any tests, and before equipment is airborne, the control knob on the height indicator shall be rotated clockwise from the OFF position and the equipment allowed five minutes to stabilize.

4.6.2.1 Comparison with Standard. The equipment shall be compared with a suitable standard during flight at the altitudes and accuracy limits listed in Table IX.

4.6.2.2 Antenna Beamwidth. At 1,000 feet, bank the aircraft 30 degrees. The height indicator shall continue to indicate 1,000 ± 45 feet.

4.6.2.3 Low Level Warning Operation. Check the operation of the low level warning by adjusting the low level index for 800 feet. As the aircraft descends, the LOW warning lamp shall light as the pointer passes directly adjacent to the index. As the aircraft ascends, the lamp shall go out before the pointer travels more than 55 feet above the index.

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4.6.2.4 Maximum Altitude Acquisition. Increase the altitude of the aircraft to greater than the maximum altitude measuring capability of the system. The indicator pointer shall remain behind the NO TRACK mask and the OFF flag shall be visible. Allow the aircraft to descend while observing the pointer. The pointer shall begin to move out from behind the mask before the aircraft descends below the maximum altitude range of the system. The OFF flag shall disappear from view as the pointer begins to move, indicating that a tracking condition has been established.

4.6.2.5 Push-To-Test Operation. Proper operation of the system may be checked during flight by pressing the control knob on the height indicator to actuate the push-to-test condition. The indicated altitude shall be 100 ± 15 feet if the system is operating properly. Releasing the control knob restores the system to normal operation. When the aircraft is at an altitude between 75 feet and 125 feet, the push-to-test condition may not function because the receiver-transmitter will not transfer track from the actual altitude to the push-to-test altitude.

4.6.2.6 Zero Reference. Note the height indicator readings as the aircraft lands. The indicated altitude at touchdown shall be zero ± 5 feet.

CAUTION: Do not remove system power when the aircraft ascends to an altitude above the Electronic Altimeter Set maximum altitude. If system power is removed and reapplied, an additional three minutes of warmup time will be required before the system can again provide accurate altitude information.

5.0 PREPARATION FOR DELIVERY

This section is not applicable to this specification.

6.0 NOTES

6.1 Intended Use. The Electronic Altimeter Set is intended for use during low level operation to sense absolute altitudes of zero to 5000 feet over the terrain. It is not affected by atmospheric or barometric conditions and provides precise altitude information down to zero feet of altitude for use during aircraft landings. The system employs narrow pulse transmission and leading-edge tracking of the echo return pulse to provide a continuous indication of aircraft altitude. Actual altitude is displayed on a servo-actuated height indicator. Signals are provided to external systems to enable monitoring, telemetering, or automatic flight control of aircraft altitude.

6.2 Zero Altitude Calibrations. The Electronic Altimeter Set is essentially a time measuring device, that is, it measures the time between the transmitted and received signal. The antenna cables, antenna height and antenna spacing affect this time measurement. The effect of these parameters are offset for a particular installation by one adjustable resistor within the receiver-transmitter.

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UNSERVICEABLE NEW MATERIAL REPORT
(FOR UNSERVICEABLE NEW MATERIAL OR MATERIAL NOT MEETING DRAWINGS AND SPECIFICATIONS)
(AMCR 715-60)

1. DATE

2. TO:

3. FROM:

4. ITEM *(Name, Drawing Number and Serial Number)*

5. MANUFACTURER OR SHIPPER

6. FOR USE ON END ITEM

I. AVAILABLE REFERENCES

7. CONTRACT NUMBER	10. PRODUCTION ORDER NUMBER	13. SHIPPING POINT	16. SHIPPING TICKET
8. SHIPPING ORDER	11. B/L NUMBER	14. LOT NUMBER	17. DATE PACKED
9. DATE RECEIVED	12. QUANTITY	15. INSPECTION REQUISITION NO.	18. FROM

II. RESULTS OF INSPECTION AND/OR ASSEMBLY

19. TOTAL INSPECTED	21. NUMBER FOUND FAULTY	23. PERCENT FOUND FAULTY
20. TRIED IN ASSEMBLY	22. NUMBER FAILED	24. PERCENT FAILED

III. CONDITION OF MATERIAL AND REASON FOR MATERIAL BEING UNSATISFACTORY

25.

IV. DISPOSITION RECOMMENDED

26. USABLE ☐ YES ☐ NO

27. RECOMMENDATIONS ☐ REPLACEMENT ☐ CORRECTION AND REINSPECTION BY GOVERNMENT
☐ CORRECTION AND REINSPECTION BY MFG ☐ STOP SHIPMENT ☐ RETURN TO MANUFACTURER
☐ CORRECT PRODUCTION ☐ OTHER *(Specify)*

PREPARED BY

28. TYPED NAME

29. SIGNATURE

ACTION TAKEN BY

30. TYPED NAME

31. SIGNATURE

DISTRIBUTION

32.

AMC FORM 1229 13 MAY 64 SEE INSTRUCTIONS ON REVERSE SIDE

FIGURE 1. Sample of AMC Form 1229, Unserviceable New Material Report

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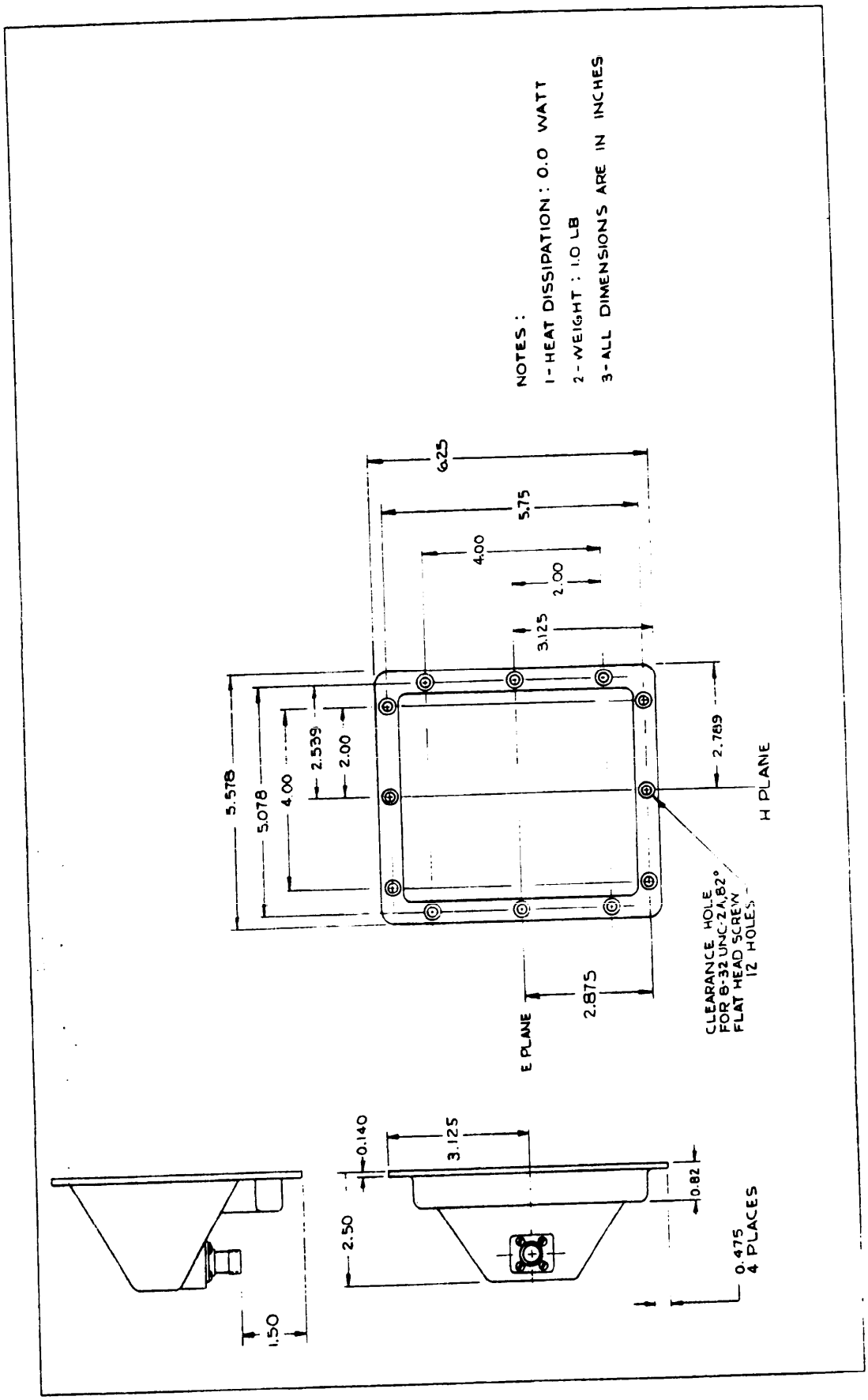


FIGURE 2. AS-1858/APN-171(V) Antenna, Outline Dimensions

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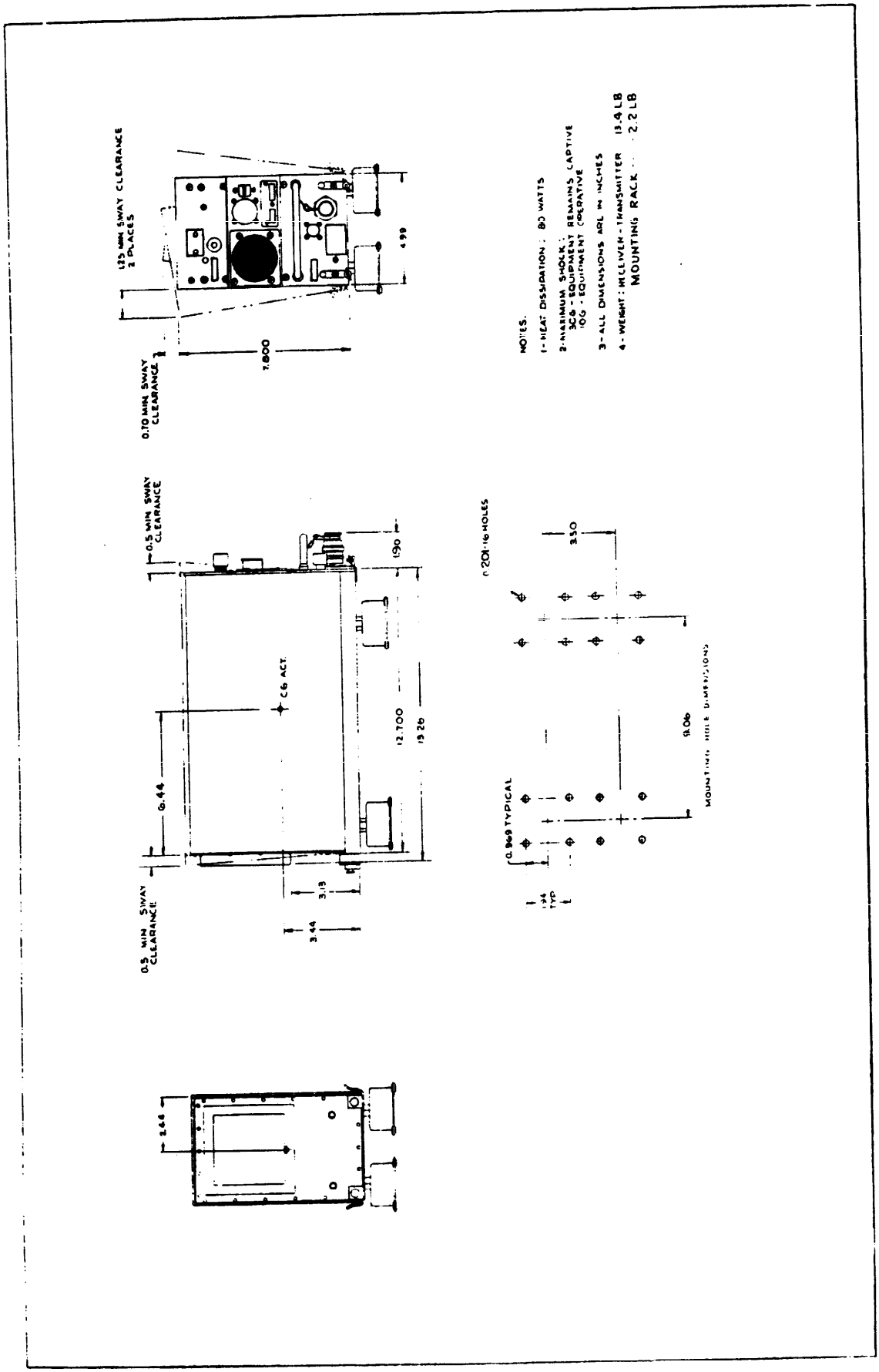


Figure 3. RT-804A/APN-171(V) Receiver-Transmitter on MT-4167/APN-171(V) Mounting Base, Outline Dimensions

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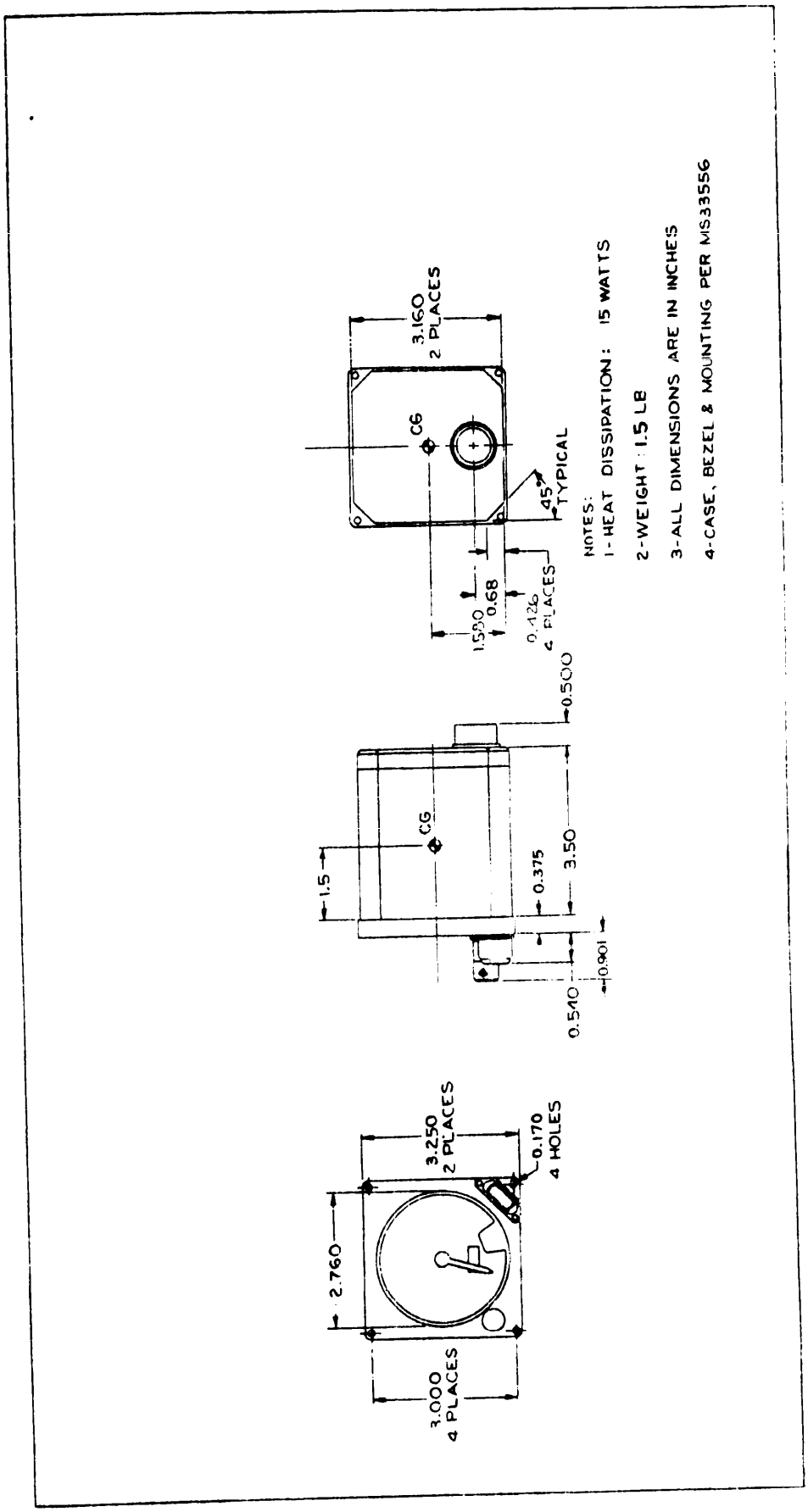


FIGURE 4. ID-1345A/APN-171(V) Height Indicator, Outline Dimensions

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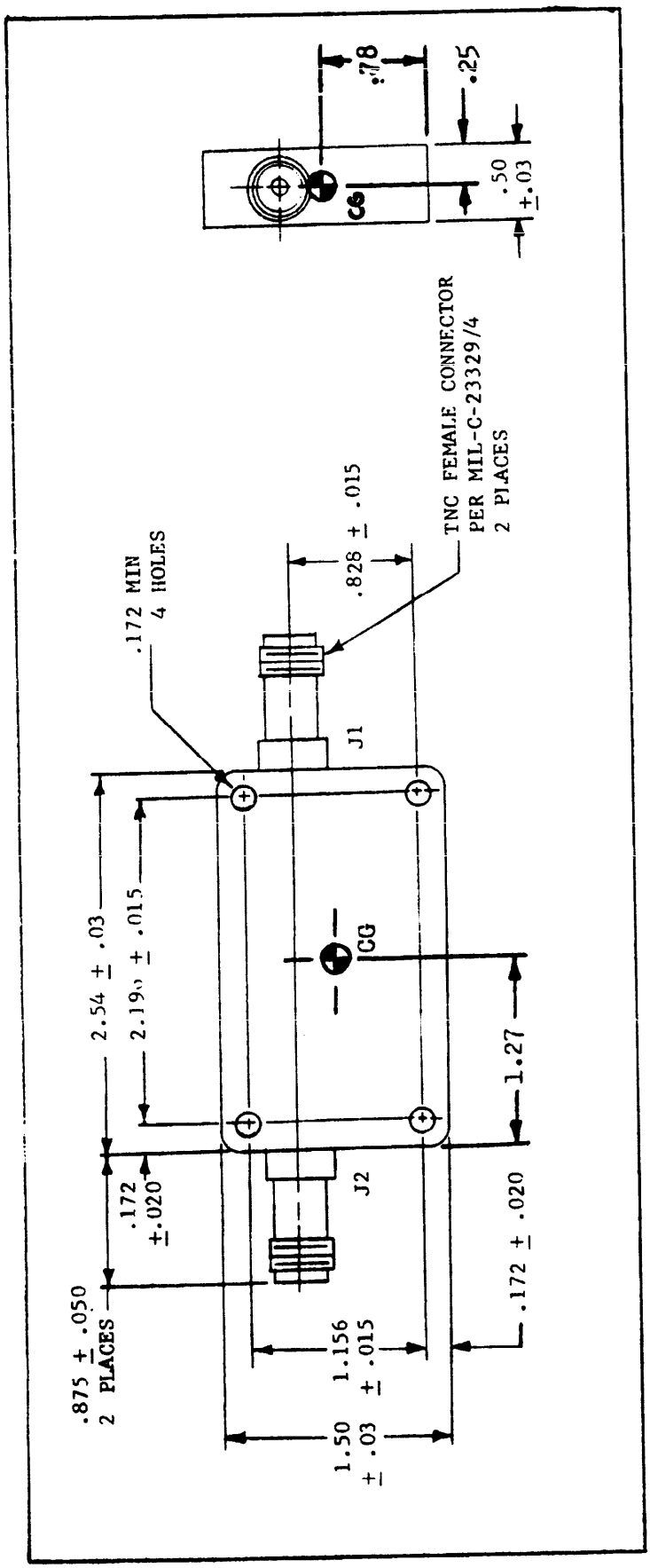


FIGURE 5. F-1263/APN-171A(V)1 Radio Interference Filter, Outline Dimensions

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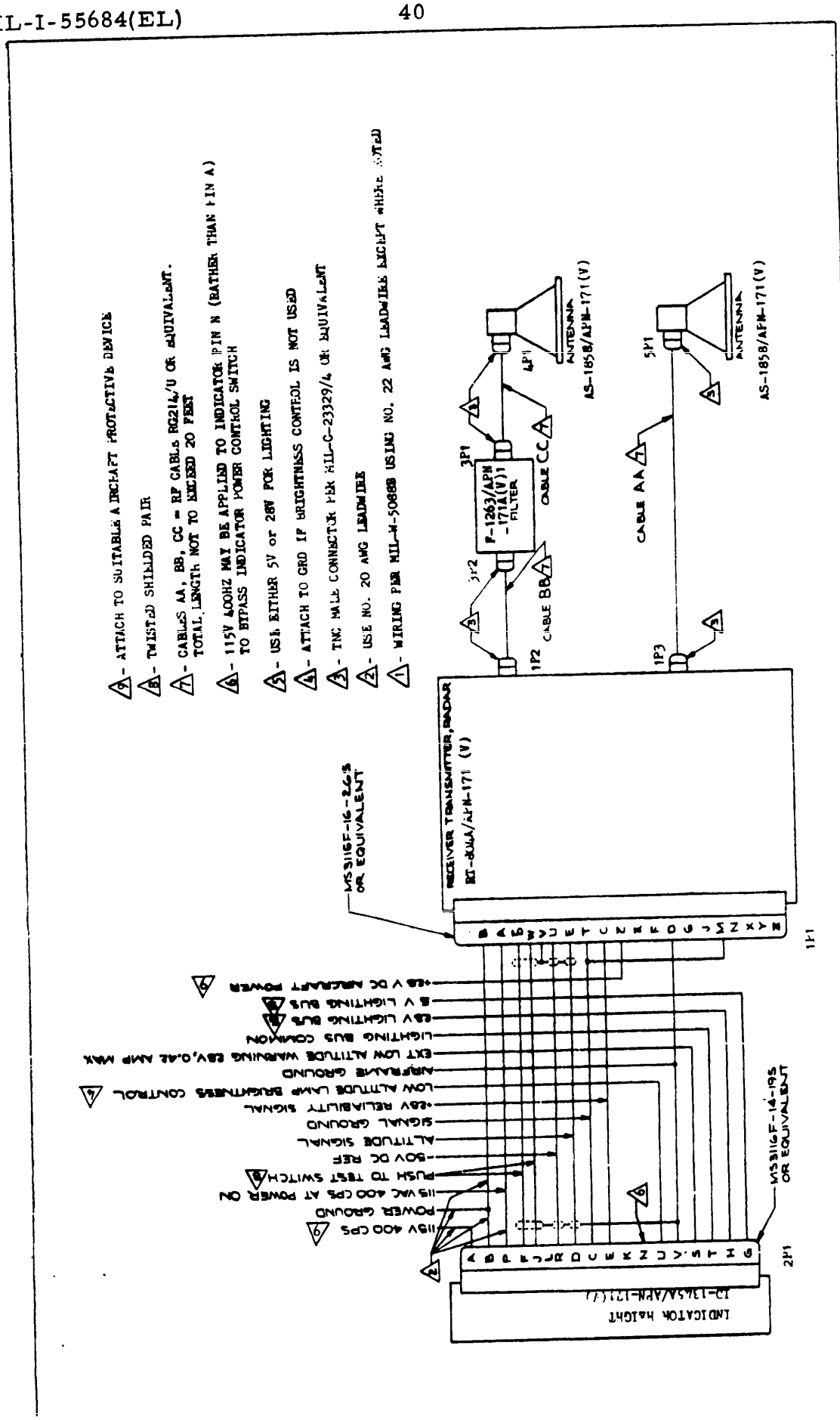


Figure 6. AN/APN-171A(V)1 Electronic Altimeter Set, External Wiring

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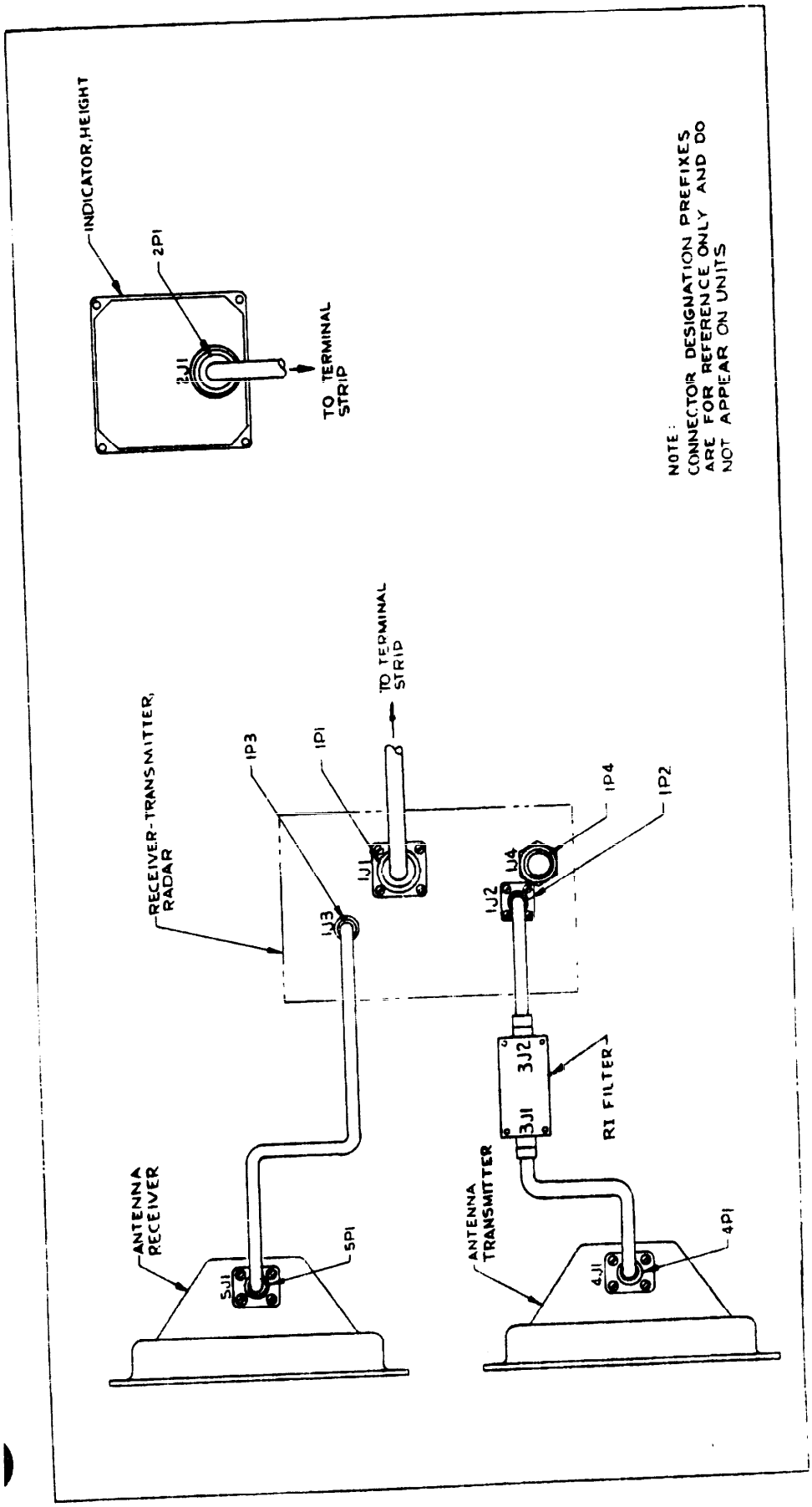


FIGURE 7. AN/APN-171A(V)1 Electronic Altimeter Set, Cabling Diagram

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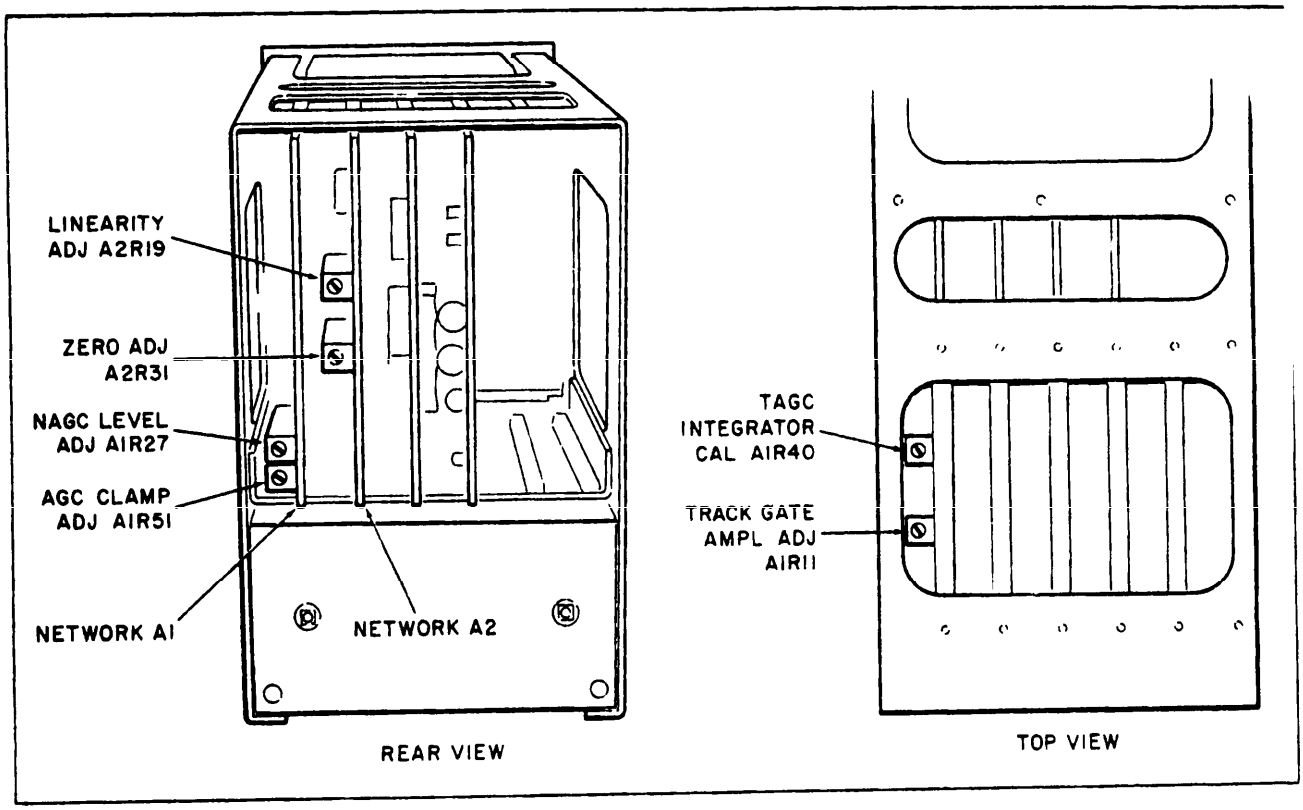
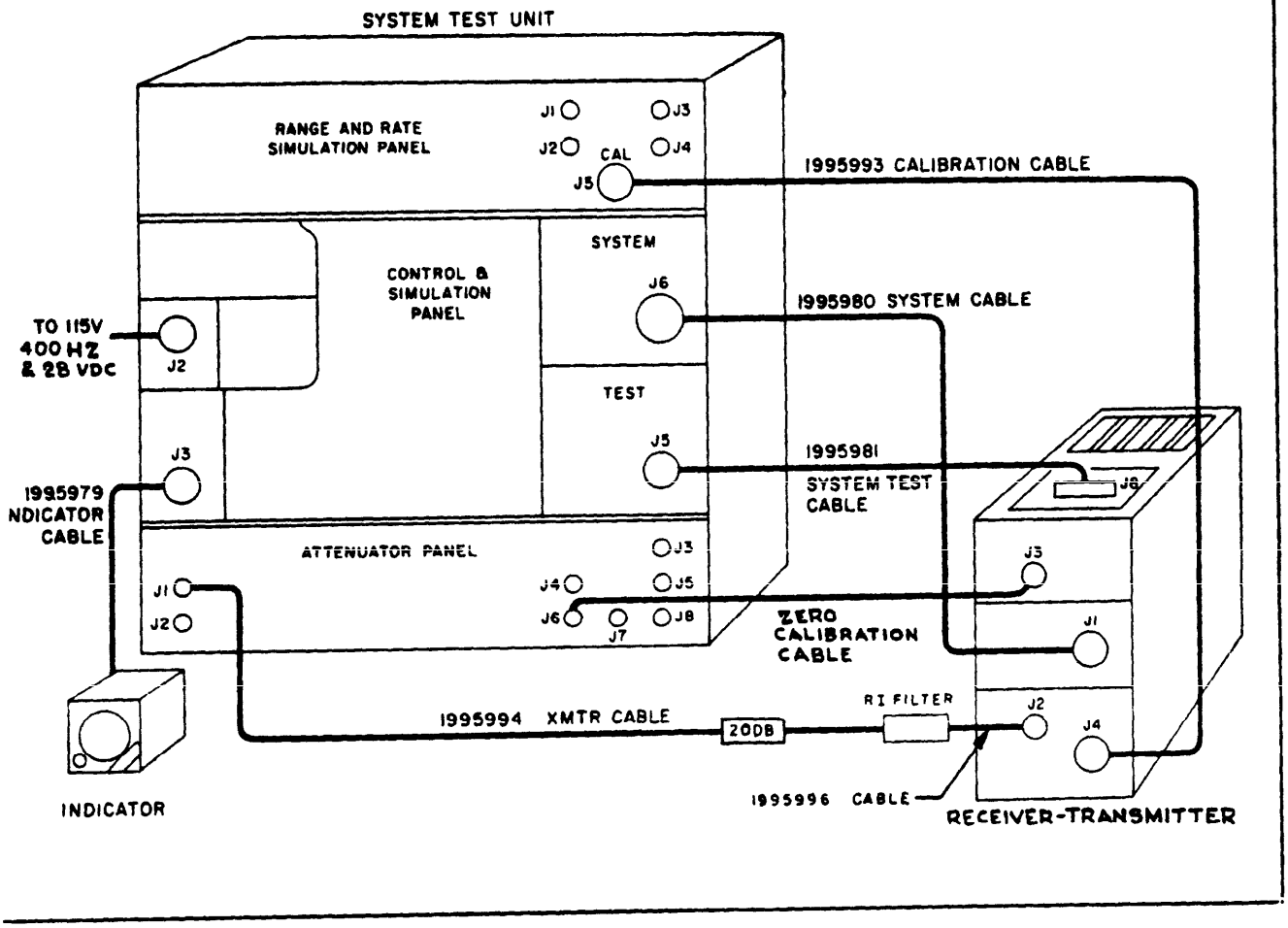


FIGURE 8. RANGE COMPUTER ADJUSTMENTS

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IGURE 9. UG2308AA01 TESTER AND ELECTRONIC ALTIMETER SET CONNECTED FOR PERFORMANCE TESTS

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APPENDIX A. CHECK-OFF LIST FOR TEST OF ELECTRONIC ALTIMETER SET, AN/APN-171A(V)1

Receiver-Transmitter, RT-804A/APN-171(V)	Serial No. _____
Height Indicator, ID-1345A/APN-171(V)	Serial No. _____
Antenna, AS-1858/APN-171(V)	Serial No. _____
Antenna, AS-1858/APN-171(V)	Serial No. _____
Mounting Base, MT-4167/APN-171(V)	Serial No. _____
Radio Interference Filter, F-1263/APN-171A(V)1	Serial No. _____

EQUIPMENT PERFORMANCE TESTS

APPLICABLE PARAGRAPH	TESTS	MINIMUM STANDARD	TEST DATA	REMARKS
4.5.1	INPUT POWER			
	400 Hz	111 to 119v		
	DC	25.0 to 28.5v		
4.5.5	INITIAL OPERATION			
	Blower	on		
	-50 vdc	-50 ± 3v		
4.5.6	POWER SUPPLY VOLTAGES	Volts DC		
	Test Jack 3	114 to 126		
	4	79 to 81		
	5	24 to 32		
	18	14.85 to 15.15		
	11	-6.23 to -6.37		
	19	-14.85 to -15.15		
	10	-49.87 to -50.13		
4.5.7	TRACK INDICATION			
	Track light	lighted		
4.5.8	ZERO ALTITUDE CALIBRATION			
	Ext. range voltage	zero ± 0.007 vdc		

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APPLICABLE PARAGRAPH	TESTS	MINIMUM STANDARD	TEST DATA	REMARKS
4.5.9	ZERO ALTITUDE ACQUISITION			
	Forced SRCH: Track light	out		
	Ext. Range Voltage	-46.7 ± 1.0 vdc		
	Norm: Track light	lighted		
4.5.10	ALTITUDE TRACK SENSITIVITY	78 ± 3db		
4.5.11	ZERO ALTITUDE ACQUISITION SENSITIVITY	75 ± 4db		
4.5.12	PUSH-TO-TEST OPERATION			
	Ext. Range Voltage	-0.80 ± 0.12 vdc		
	Indicator	100 ± 15 ft.		
4.5.13	ALTITUDE ACCURACY TESTS			
	Altitude	Volts DC		
	0 feet	0 ± 0.012		
	250 feet	-2.0 ± 0.032		
	500 feet	-4.0 ± 0.052		
	1000 feet	-8.0 ± 0.092		
	2500 feet	-20.0 ± 0.212		
	5000 feet	-40.0 ± 0.412		
4.5.14	FAR RANGE PUSH-TO-TEST OPERATION			
	Ext. Range Voltage	-0.80±0.12		
4.5.15	SCALE ACCURACY TEST			
	Scale Accuracy Altitude	Volts DC		
	0 feet	0.000 to 0.012		
	150 feet	-1.194 to -1.213		
	500 feet	-3.920 to -4.080		
	3000 feet	-23.528 to -24.471		

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APPLICABLE PARAGRAPH	TEST	MINIMUM STANDARD	TEST DATA	REMARKS
4.5.15 (Continued)				
	Repeatability Altitude	Volts DC		
	0 feet	± 0.101		
	150 feet	± 0.010		
	500 feet	± 0.020		
	3000 feet	± 0.140		
4.5.16	LOSS OF TRACK TEST			
	Indicator pointer	behind mask		
	Flag position	no track		
4.5.17	LOW LEVEL WARNING TEST			
	LAMP LIGHTS			
	Index Setting	Volts DC		
	50 feet	-0.368 to -0.432		
	150 feet	-1.168 to -1.232		
	600 feet	-4.640 to -4.960		
	3000 feet	-23.144 to -24.860		
	LAMP GOES OUT			
	Index Setting			
	50 feet	-0.400 to -0.480		
	150 feet	-1.200 to -1.280		
	600 feet	-4.800 to -5.176		
	3000 feet	-24.000 to -26.136		
4.5.18	RESPONSE TIME	2 seconds max.		

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ENGINEERING APPROVAL TESTS
INSTALLED IN _____ AIRCRAFT TAIL NO. _____

APPLICABLE PARAGRAPH	TEST	MINIMUM STANDARD	TEST DATA	REMARKS
	PREFLIGHT			
4.6.1.1	Power Sources			
	ac voltage	107 to 119v		
	dc voltage	25 to 28.5v		
4.6.1.2	Initial Operation			
	Off flag	disappears within 3 min.		
4.6.1.3	Indicator Illumination	lighted		
4.6.1.4	Warning Lamp and Push-to-test operation			
	Altitude	100 ± 15 feet		
	Low warning lamp during push-to-test	0 feet		
	Low warning lamp after push-to-test	lighted		
	FLIGHT			
4.6.2.1	Comparison with Standard			
	Altitude	Indication (feet)		
	50 feet	41.9 to 58.1		
	100 feet	90.4 to 109.6		
	190 feet	177.6 to 202.4		
	1000 feet	955 to 1045		
	5000 feet	4768 to 5232		
4.6.2.2	Antenna Beamwidth			
	30 degree bank			

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APPLICABLE PARAGRAPH	TEST	MINIMUM STANDARD	TEST DATA	REMARKS
4.6.2.3	Low Level Warning Operation			
	Low lamp lights	at index		
	Low lamp goes out	within 55 ft. above index		
4.6.2.4	Maximum Altitude Acquisition			
	Above Range			
	Indicator pointer	behind mask		
	Off flag	visible		
	Re Gains Track	no less than 5000 feet		
4.6.2.5	Push-to-test Operation	100 ± 15 feet		
4.6.2.6	Zero Reference			
	Height indication	0 ± 5 feet		

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PRODUCTION APPROVAL TESTS
INSTALLED IN _____ AIRCRAFT TAIL NO. _____

APPLICABLE PARAGRAPH	TEST	MINIMUM STANDARD	TEST DATA	REMARKS
	PREFLIGHT			
4.6.1.2	Initial Operation			
	Off flag	disappears within 3 min.		
4.6.1.3	Indicator Illumination	lighted		
4.6.1.4	Warning Lamp and Push-to-test operation			
	Altitude	100 ± 15 feet		
	Low warning lamp during push-to-test	0 feet		
	Low warning lamp after push-to-test	lighted		
	FLIGHT			
4.6.2.2	Antenna Beamwidth			
	30 degree bank	955 to 1045 ft.		
4.6.2.3	Low Level Warning Operation			
	Low lamp lights	at index		
	Low lamp goes out	within 55 ft. above index		
4.6.2.4	Maximum Altitude Acquisition			
	Above Range			
	Indicator Pointer	behind mask		
	Off flag	visible		

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APPLICABLE PARAGRAPH	TEST	MINIMUM STANDARD	TEST DATA	REMARKS
	Re Gains Track	no less than 5000 feet		
4.6.2.5	Push-to-test Operation	100 ± 15 feet		
4.6.2.6	Zero Reference			
	Height indication	0 ± 5 feet		

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ARMY-EL
Project No. 5841-A030

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-I-55684(EL), INSTALLATION AND ACCEPTANCE TESTING OF ELECTRONIC ALTIMETER SET AN/APN-171A(V)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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